

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. 8389

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	3DS	Seat No:	2022700045
Course:	Section:1				

L_41_658_986_1_1505_41376



Name of the candidate. Parvati Pole

Candidate's UID number:

2	0	2	2	7	0	0	0	4	5
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Examination class room number:

3	0	5
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Invigilator's signature with date:

	8/10
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(To be filled in by the candidate)

EXAMINATION: TY BTECH
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: CSDS

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 08.10.2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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(Begin answer for each question on a new page)

1. a) $f(w) = \frac{1}{2} \|xw - y\|^2$

$$\frac{\partial f(w)}{\partial x} = \frac{2}{2} \|xw - y\|$$

$$\frac{\partial f(w)}{\partial x} = \|xw - y\| \times w$$

$$\frac{\partial f(w)}{\partial x} = w \|xw - y\|$$

$$\frac{\partial f(w)}{\partial x} = w \|xw - y\|$$

$$= xw^2 - y$$

$$\frac{\partial f(w)}{\partial y} = \frac{2}{2} \times \|xw - y\| \times (-1)$$

$$\frac{\partial f(w)}{\partial y} = -\|xw - y\|$$

⇒ Hebbian matrix = $\begin{vmatrix} \frac{\partial f(w)}{\partial x} & \frac{\partial f(w)}{\partial y} \\ \frac{\partial f(w)}{\partial x} & \frac{\partial f(w)}{\partial y} \end{vmatrix}$

gradient update rule = $w_0 = \eta \frac{\partial J}{\partial w}$

⇒ $w(J) = w_0 - \eta \frac{\partial J}{\partial w}$

⇒ ~~$X^T X$~~ ~~max~~ eigen value of ~~$X^T X$~~

So, $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a^2 + b^2 + c^2 +$

⇒ AH

~~w_0~~ $w_0 \cdot x \leq 0$ and $w_0 \cdot x > 0$

Problem :- If P and N are two linearly separable subsets belongs to $\{x_1, x_2, x_3, \dots, x_m\}$ then convergence point should be possible $\in \{x_1, x_2, x_3, \dots, x_m\}$

preposition :- for P $w_0 \cdot x_1 \geq 0$ and for N it should be $w_0 \cdot x_2 \leq 0$

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$$1. b) \text{MSE} = (y - \hat{y})^2$$

$$\hat{y} \text{ lets assume } = \text{mix} + c$$

$$\text{MSE} = (y - (\text{mix} + c))^2$$

$$= y - (m^2x^2 + c^2 + 2mixc)$$

$$= y - (m^2x^2 + 2mixc + c^2)$$

$$\Rightarrow y - \text{mix}(\text{mix} + 2c + c^2)$$

$$\Rightarrow y = \text{mix}(\text{mix} + c(2+c))$$

Now, see overfitting occurs when model performs ~~poor~~ gives poor performance on training and testing data both. This happens due to the noise present in the dataset. Model learns extra parameters, noise which affects model performance. To reduce this problem we need to reduce unnecessary parameters when model capacity is too high, we can expect high number of parameters.

$$y_{\text{out}} = w_0x + b_0 \quad b_0 = \text{bias.}$$

$$\text{MSE} = (y_{\text{expected}} - \hat{y}_{\text{actual}})^2$$

$$= (y - (w_0x + b_0))^2$$

$$\text{MSE} = (y - (w_0^2x^2 + b_0^2 + 2w_0x))$$

$$\text{MSE} = (y - (w_0^2x^2 + 2w_0x + b_0^2))$$

$\Rightarrow$ So, To minimize the error

$$\frac{\partial(M)}{\partial x} = -\cancel{w_0^2} + 2w_0 \quad \frac{\partial(M)}{\partial y} =$$

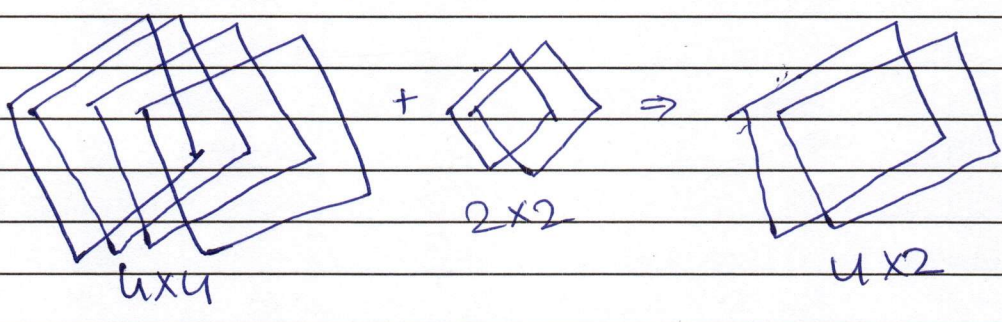
$$= w_0(-w_0 - 2)$$

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2. a) input size = $H \times W \times C$ Filter = K
 size = $F \times F$
 size of input = ~~$C \times K$~~ $F \times F$ to calculate
 no. of filter size.
 size of : $\Rightarrow F \times F \times C$
 Total no. of $\Rightarrow K \times F \times F \times C$ ✓
 parameters
 So, number of multiplications required
 will be $\Rightarrow 3$.

$\Rightarrow D_{input} = C$ $D_{output} = K$
 filter size = F
 $\Rightarrow$ standard 2D convolution.
 $\Rightarrow C \times F \times F$ -- [to calculate number of input
 size]
 $\Rightarrow K \times C \times F \times F$ [for after filter applied]
 Depthwise separable convolution



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2. b) Autoencoder is like a coder who will encode parameters.

⇒ Overcomplete autoencoder tends to learn the identity function because its PCA ~~is like~~ does feature extraction through to reduce the parameters to avoid overfitting and underfitting issues.

⇒



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3.d) gradient expression = $\frac{\partial L}{\partial h_{t-k}}$

~~forget gate LSTM~~ → Recurrent neural networks acts as a short memory model. It can't remember data long time. While LSTM is like large memory model who can remember the things after long time also. For example RNN is like student having poor memory who can't remember everything and LSTM acts like a brilliant student who remember all the chapters.

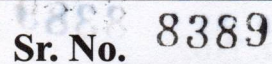
forget gate :- forget gate of LSTM will forget unnecessary information or the information which is not required or not used recently. Useful.

This is the feature which makes LSTM exceptional than the RNN.

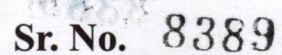
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8. b) GRU is the version of LSTM only. we can say that GRU is the type of LSTM
⇒ - GRU :-

[illegible]

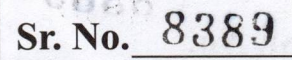
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[illegible]

Question No.	Answer
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[illegible]

[illegible]

~~8389~~

[illegible]

[illegible]

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Rough work

 $62 \times 62 \times 5$, filter 3×3

$$3 \times 3 = 9$$

$$9 \times 5 = 25$$

$$25 + 9 =$$